

PROJECT GUIDE & ARCHITECTURE

Basal Cell Carcinoma Skin Detection Using Deep Learning

A Comprehensive Step-by-Step Explanation of Data Ingestion, Custom SE-STN-EfficientNet Modeling, Optimization, Calibration, and Explainable AI.

Target Domain: Clinical Dermatology & Computer-Aided Diagnosis (CAD)

Aggregated Datasets: HAM10000, ISIC2019, PAD-UFES-20

Deep Learning Backbone: EfficientNet (B5 / B0)

Custom Blocks: Squeeze-and-Excitation (SE), Spatial Transformer Network (STN)

XAI Suite: Grad-CAM, Grad-CAM++, Score-CAM, Eigen-CAM, Integrated Gradients

Step 1: Dataset Aggregation & Curated Splits

The system is trained to solve both binary classification (BCC vs. non-BCC) and multi-class subtype diagnostics. The dataset is consolidated from three major clinical repositories:

- **HAM10000**: Contains 10,015 dermoscopic images representing common pigmented lesions.
- **ISIC2019**: Contains 25,331 high-resolution images, heavily utilized for skin cancer classification benchmarks.
- **PAD-UFES-20**: A clinical skin cancer dataset captured using smartphone cameras, introducing realistic out-of-distribution noise.

The aggregated data is structured into csv tables (train.csv, val.csv, test.csv) containing image paths, clinical labels, diagnosis subtypes, and Fitzpatrick skin phototype classifications (grades I-VI).

Step 2: Preprocessing & Augmentation Pipeline

To ensure the model learns legitimate dermatological features instead of artifacts (such as hairs, air bubbles, or varying illumination conditions), images pass through an advanced preprocessing pipeline:

Dermoscopic Hair Removal

The hair removal routine uses morphological filtering to eliminate occlusion. First, the image is converted to grayscale and blurred. A **Blackhat morphological filter** is applied using an elliptical kernel, extracting narrow, dark hair structures. The extracted structures are thresholded and dilated to form a binary mask. Finally, **Fast Marching / Navier-Stokes Inpainting** (cv2.inpaint) fills in the hair pixels using the color values of neighboring skin pixels, producing a clear, hair-free lesion.

Reinhard Color Normalisation

Camera hardware and lightning variations introduce color distribution shifts. Reinhard normalization transforms the BGR image into the **CIE L*a*b* color space** (which decouples luminance from color channels). The mean and standard deviation of the L, a, and b channels are computed and scaled to match a pre-defined healthy target skin template. This removes camera/lighting bias while retaining lesion-specific color changes.

Offline Preprocessing & Online Augmentations

To avoid CPU bottlenecks during training, a multiprocessing script batch-preprocesses all images offline. During online training, the DataLoader applies standard augmentations (Horizontal/Vertical Flips, Rotations, Color Jitter, Gaussian Blur) and specialized **minority class augmentations** (Elastic Transformations, Grid Distortions, Shift-Scale-Rotate) exclusively to BCC samples to counteract class imbalance.

Step 3: Neural Network Architecture (SE-STN-EfficientNet)

The architecture incorporates a standard CNN backbone alongside structural attention and spatial normalization blocks:

1. Feature Backbone

A pretrained EfficientNet convolutional network (efficientnet_b5 or a lightweight efficientnet_b0) acts as the feature extractor, capturing multi-scale visual details from its mid-level MBConv stages.

2. Channel Squeeze-and-Excitation (SE)

An SE Block is appended after each backbone stage. It performs global average pooling to compress spatial dimensions into a channel descriptor. Two fully connected layers compute channel weights, which are scaled back onto the feature maps. This dynamically emphasizes informative feature maps while suppressing redundant ones.

3. Spatial Transformer Network (STN)

Before global pooling, the deepest feature map passes through an STN module. A localization network learns to predict a 6-parameter affine transformation matrix. By applying grid sampling, the STN dynamically rotates, translates, and crops the feature map. This **geometrically normalizes the lesion**, focusing the head on the anomaly regardless of rotation, zoom, or camera angle.

4. Classification Head

Features are pooled via Global Average Pooling and passed to a three-layer fully connected head (1024 -> 512 -> classes) with GELU activations, Batch Normalization, and Dropout to prevent overfitting.

Step 4: Advanced Training Protocol

The training loop employs a rigorous optimization pipeline designed to maximize generalization:

- **Discriminative Learning Rates:** The backbone parameters are trained at a 10x lower rate than the SE, STN, and Head parameters, protecting pretrained weights from disruptive gradients.
- **Hybrid Focal Cross-Entropy Loss:** Combines standard CE loss with Focal Loss. This addresses class imbalance by dynamically weighting samples, forcing the model to focus on hard, misclassified samples.
- **CutMix Augmentation:** Random patches of two training images are merged, and their target labels are proportionally blended, forcing the model to focus on global patterns instead of single local features.
- **EMA & SWA:** Exponential Moving Average weights are maintained for stable evaluation. In final epochs, Stochastic Weight Averaging averages checkpoints to find flat minima that resist overfitting.

Step 5: Post-Training Calibration & Threshold Tuning

Medical diagnostic models require calibrated risk assessments and optimized sensitivity boundaries:

Temperature Scaling Calibration

Deep learning models are notoriously overconfident. The project includes a post-hoc calibration step using Temperature Scaling. By evaluating the model on a validation set, an L-BFGS optimizer fits a single scaling parameter T to minimize negative log-likelihood. During inference, raw logits are scaled as logit/T , minimizing Expected Calibration Error (ECE) and aligning the model's confidence scores with real-world accuracy.

Clinical Threshold Optimization

Missing a malignant lesion (false negative) is a dangerous diagnostic error. To maximize clinical safety, the pipeline sweeps the precision-recall curve on the validation set, choosing an optimal decision boundary of **0.40** (instead of the standard 0.50). This increases model sensitivity (recall) to capture early-stage Basal Cell Carcinomas.

Step 6: Clinical Diagnostics & Explainable AI (XAI) Suite

To assist dermatologists and build clinical trust, the model provides visual interpretations and safety analytics:

Attribution Visualizations

The system implements a multi-method XAI engine, targeting the final convolutional layers:

- **Grad-CAM:** Pools gradients to highlight the regional features most responsible for the diagnosis.
- **Grad-CAM++:** Uses higher-order derivatives to provide sharper, cleaner localization of multiple target regions.
- **Score-CAM:** A gradient-free method that projects activation maps onto the input image, measuring output score changes (optimized using the top 16 active channels).
- **Eigen-CAM:** Applies Principal Component Analysis (PCA) directly to deep activations, bypassing gradients.
- **Integrated Gradients:** Integrates gradients along the linear path between a black image baseline and the input image, establishing pixel-level attribution.

Lesion Segmentation

An OpenCV-based segmentation routine processes raw input images. By converting to grayscale, applying Gaussian blurring, and executing Otsu's Adaptive Thresholding, it detects the boundaries of the primary lesion. It isolates the largest contour and overlays a bright clinical cyan boundary, giving the physician visual confirmation of the segmented lesion area.

Fitzpatrick Fairness Assessment

To prevent algorithmic bias across different skin tones, the evaluation pipeline computes the model's ROC-AUC separately for each Fitzpatrick skin phototype group (I to VI), ensuring equal diagnostic fidelity for all patient populations.

Step 7: Web Dashboards (Flask & FastAPI)

The project features two web dashboards:

Flask Dashboard (app.py)

A lightweight interface that accepts image uploads, performs preprocessing, runs basic inference with Test-Time Augmentation (TTA), and saves a Grad-CAM overlay image.

Clinical FastAPI Dashboard (app_fastapi.py)

An advanced, production-ready diagnostic dashboard:

- **Webcam & Upload Ingestion:** Captures live webcam feeds or accepts image file uploads.
- **Multi-Model Ensemble & TTA:** Runs predictions across B0 and B5 models, combining them with Test-Time Augmentations (original, horizontal flip, vertical flip) for enhanced stability.
- **9-Class Disease Spread:** Extrapolates the binary prediction to 9 clinical categories (Melanoma, Benign Nevus, etc.) using realistic prior probabilities.

- **Full Diagnostics:** Renders the preprocessed skin image, the segmented lesion boundary, and 4 CAM heatmaps simultaneously.
- **SQLite Logging:** Stores records containing patient details (name, age, gender), model metadata, doctor notes, and visual output paths in a database. Records can be searched, deleted, or exported directly to an Excel sheet.
- **Telemetry Stats:** Monitors server CPU, RAM, and GPU memory usage in real time.